Querying and Analyzing SQL Server 2025

A three-day course in writing, analyzing and optimizing Transact-SQL queries on SQL Server 2025 with SQL Server Management Studio 22. Many lessons also apply to Azure SQL Database environments.

DURATION	CONTENT	LEVEL	DELIVERY
3 Days	10 Modules	Beginner	Instructor-Led · Bootcamp · Self-Paced

WHAT SETS THIS COURSE APART

01

Updated for SQL Server 2025

Queries are written, analyzed and tuned on SQL Server 2025 using SQL Server Management Studio 22.

- Native JSON and JSON indexes
- Intelligent Query Processing and Optimized Locking
- SQL Server 2025 string enhancements

02

AI features in the engine and tools

Students use vector search in the database engine and AI assistance in SSMS 22.

- Vector similarity search in SQL Server 2025
- GitHub Copilot in SSMS 22 for query assistance
- AI-powered execution plan analysis

03

Lab simulations, at the student's pace

Students step back and forth through each lab exercise, revisiting any step as needed.

- Runs in a browser, with no setup required
- Replay any time, including after class
- Covers the lab in every module

ALSO INCLUDED

AI Tutor, 24/7	An interactive AI assistant built into the courseware platform answers questions on every course topic, at no extra cost.
Bootcamp mode	One capstone lab per day. Students apply that day's skills to a new scenario with no procedure and no hints.
Analytical queries	Window functions, subqueries and common table expressions for complex reporting.
Indexing	Clustered, nonclustered, covering and JSON index design.

CURRICULUM AT A GLANCE

DAY 1	**Query fundamentals.** SSMS 22, basic Transact-SQL and string enhancements, and filtering and sorting data.
DAY 2	**Complex and modern data.** Joins, subqueries and table expressions, window functions, and native JSON and vector search.
DAY 3	**Analysis and optimization.** Execution plans with AI, indexing strategies, and Intelligent Query Processing.

What your team leaves with

The ability to write and execute Transact-SQL queries with SSMS 22 and Copilot assistance, analyze performance using execution plans and AI-powered interpretation tools, optimize queries through indexing, and use SQL Server 2025 features including native JSON, Intelligent Query Processing and Optimized Locking.

Prerequisites: basic understanding of relational database concepts, familiarity with Windows, and basic computer navigation skills.
Next step: Course 65318A, Advanced SQL Server Performance Tuning.

COURSEWARE ACCESS

softwaretutorialservices.com/course/65317a

support@softwaretutorialservices.com